

Ameya Patil

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ACADEMIC & PROFESSIONAL INTERESTS

My academic and professional interests lie in the intersection of Databases, Data Visualization and Human Computer Interaction. I build interactive data visualization and analysis systems to cater to the growing need of big data analytics. Specifically, my focus is on **network data** visualization and analysis. I am also interested in visualization perception research with regard to uncertainty visualization, and my passion and awareness about environmental issues drive me towards applications of my research interests in spreading literacy through data visualizations, especially in the space of **socio-environmental** data science contexts.

EDUCATION

University of Washington, Seattle (UW) WA, USA
Ph.D. in Computer Science, Advised by **Dr. Leilani Battle**, GPA: 4.00/4.00 Jan 2021 - Present

- **Thesis:** Benchmarking and Designing Scalable Interactive Network Visualization and Analysis Systems
- **Relevant Coursework:** Computing for Conservation, Computer Science Education seminar, Game Design

University of Maryland, College Park (UMD) MD, USA
M.S. in Computer Science, GPA: 3.84/4.00 Aug 2018 - Dec 2020

- **Relevant Coursework:** Machine Learning, Geometric Computer Vision, Advanced Computer Graphics, Physically Based Modelling, Simulation & Animation, Interactive Data Analytics, Computational Geometry, Interactive Technologies in HCI, Database System Architecture and Implementation

Birla Institute of Technology and Science - Pilani (BITS) Goa, India
B.E. (Honors) in Computer Science, GPA: 8.24/10.00 Aug 2012 - May 2016

- **Electives:** Data Mining, Data Storage Technologies and Networks, Creative Multimedia

PUBLICATIONS

1. Z. Dong, T. Barrett, **A. Patil**, Y.Shoda, L. Battle, E. Wall, “A Design Space of Behavior Change Interventions for Responsible Data Science”, Proceedings of the Conference on Intelligent User Interface (IUI), 2025. DOI: 10.1145/3708359.3712140. [pdf](#)
2. Z. Dong, **A. Patil**, Y.Shoda, L. Battle, E. Wall, “Behavior Matters: An Alternative Perspective on Promoting Responsible Data Science”, Proceedings of the ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW), 2025. DOI: 10.1145/3710932. [arxiv:2410.17273](#)
3. **A. Patil**, M. Smith, H. Kershaw, M. El Gharamti, “Interactive Visualization of Ensemble Data Assimilation Forecasts for Freshwater Floods”, IEEE VIS Viz4Climate + Sustainability Workshop Demo, 2024. [pdf](#)
4. **A. Patil**, Z. Rand, T. Branch, L. Battle, “WhaleVis: Visualizing the History of Commercial Whaling”, IEEE VIS Short Papers, 2023. DOI: 10.1109/VIS54172.2023.00028. [arxiv:2308.04552](#)
5. **A. Patil**, Z. Rand, T. Branch, L. Battle, “WhaleVis: A New Visualization Tool for the IWC Catch Database”, International Whaling Commission SC/69A/GDR/04, 2023. archive.iwc.int/SC/69A/GDR/04
6. **A. Patil**, G. Richer, C. Jermaine, D. Moritz, J.-D. Fekete, “Studying Early Decision Making for Progressive Bar Charts”, IEEE Transactions on Visualization and Computer Graphics, 2023. DOI: 10.1109/TVCG.2022.3209426. HAL: <https://hal.science/hal-03738461/>

7. A. Aguinaldo, P.-Y. Chiang, A. Gain, **A. Patil**, K. Pearson and S. Feizi, “Compressing GANs using Knowledge Distillation”, CoRR, vol. abs/1902.00159, 2019. [arXiv:1902.00159](#)

EXPERIENCE

National Renewable Energy Laboratory
Research Intern, advised by **Dr. Sam Molnar**

Golden CO, USA
Apr 2025 - Sept 2025

- Research intern in the Data Analysis and Visualization Group (DAV), working on GPU based network data visualization

National Center for Atmospheric Research
Data Visualization Intern, advised by **Marlee Smith, Dr. Helen Kershaw**
and **Dr. Moha El Gharamti**

Boulder CO, USA
Summer 2023

- Designed HydroVis - an interactive analysis dashboard for the WRF-Hydro hydrological forecasting model
- Implemented the dashboard backend using Python-Flask and XArray, and the renderer using D3.js
- Integrated the dashboard with NCAR’s Data Assimilation Research Testbed (DART) tool

University of Washington
PhD Student, advised by **Dr. Leilani Battle**

Seattle WA, USA
Feb 2022 - April 2023

- Performed a design study of domain experts in whale research to gather analysis requirements
- Designed WhaleVis - an interactive analysis dashboard for historical whale hunting data
- Implemented the dashboard backend using DuckDB, and the renderer using D3.js in Observable notebook
- Published work in IEEE VIS 2023 and presented the same at the International Whaling Commission Scientific Committee Meeting 2023, Bled, Slovenia and IEEE VIS 2023, Naarm, Australia

AVIZ, Inria
Research Intern, advised by **Dr. Jean-Daniel Fekete**

Saclay, Paris, France
Summer 2021

- Worked on understanding the efficacy of confidence intervals for decision making using progressive bar charts
- Proposed and studied the efficacy of two new visualization designs for progressive bar charts
- Studied the performance of humans vs automated statistical test for the task of answering questions based on progressive visualizations
- Published work in TVCG 2023 and presented the same at IEEE InfoVIS 2022, Oklahoma City, USA

Fraunhofer CESE
Research Assistant Intern, advised by **Dr. Marcel Schäfer**

College Park MD, USA
Summer 2019

- Worked as Java developer on the [PocketSecurity](#) project which collects data to perform user behaviour analysis
- Identified and implemented critical data probes to be collected for better analysis and improved existing probes

NVIDIA
System Software Engineer - C/C++

Pune MH, India
July 2016 - July 2018

- Worked as developer for Shadowplay - a gameplay sharing app to record, screenshot, broadcast and coplay video games
- Worked on multi-threaded and multi-processes features, GPU driver code and render pipeline
- Enhanced and monitored the automated software testing suite and guided an intern for the same

NVIDIA
Intern

Pune MH, India
July 2015 - Dec 2015

- Device Filter Drivers - C/C++: Implemented end-to-end user input redirection from input devices to a specific application using filter drivers and device notifications
- Z-buffer - Python: Implemented aesthetic visual effects such as zoom burst using the depth data of images

TALKS

- Lightning Talk for “HydroVis: Interactive Visualization of Ensemble Data Assimilation Forecasts for Freshwater Floods“ (virtual talk) - [video](#) St.Pete Beach, FL, USA
IEEE InfoVIS Viz 4 Climate + Sustainability Workshop Oct 2024
- Presented thesis proposal (virtual talk) St.Pete Beach, FL, USA
IEEE InfoVIS Doctoral Colloquium Oct 2024
- Presented thesis proposal Seattle, WA, USA
University of Washington, Seattle - DUB Doctoral Colloquium May 2024
- Presented “WhaleVis: Visualizing the History of Commercial Whaling” - [video](#) Naarm, Australia
IEEE InfoVIS Oct 2023
- Presented “HydroVis: Decision Support System for Ensemble Data Assimilation Forecasts of Freshwater Floods“ - [video](#) Boulder, CO, USA
National Center for Atmospheric Research (NCAR) Aug 2023
- Presented “Studying Early Decision Making for Progressive Bar Charts” - [video](#) Oklahoma City, OK, USA
IEEE InfoVIS Oct 2022

TEACHING

- Head Teaching Assistant** at University of Maryland, College Park Fall 2018, Fall 2019, Fall 2020
Computer Systems Architecture (CMSC411)
- Teaching Assistant** at University of Maryland, College Park Spring 2019, Spring 2020
Introduction to Data Visualization (CMSC498O)
- Teaching Assistant** at University of Maryland, College Park Spring 2021
Advanced Data Structures (CMSC420)

SKILLS

- Programming Languages:** C/C++, Java, Python, Javascript, SQL, Cypher, GSQL
- Libraries/Frameworks:** D3.js, OpenCV, MPI, OpenMP
- Miscellaneous:** DSLR Photography, Adobe Lightroom

LANGUAGES

- Marathi:** Native
- Hindi:** Fluent
- English:** Fluent

SCHOLARSHIPS AND AWARDS

- Birla Institute of Technology and Science - Pilani Merit Scholarship 2012
- Maharashtra State Board of Secondary and Higher Secondary Education Scholarship 2012

PROFESSIONAL SERVICES

- Pre-Application Mentorship Service** 2023, 2025
Guided PhD aspirants in navigating the PhD application process
- Open Source Contributions** 2025
Contributed bug-fixes and suggested features for TigerGraph Python driver and TigerGraph Graph Data Science Library

- **Peer-reviewer** 2022-2025
Reviewed submissions for TVCG 2023, SIGCHI LBW 2025, InfoVIS 2025
- **Panelist at Northwest Database Society (NWDS)** 2025
Served as a panelist on the topic of [Graph Data Management](#)
- **Graduate Application Reader** 2023
Reviewed PhD applications for University of Washington, Seattle
- **K-12 Outreach** 2022
Presented my research work to high school students during the University of Washington, Seattle K-12 outreach program

EXTRACURRICULAR ACTIVITIES

- **Social Events Chair at the University of Washington, Seattle** 2022
Organized friday evening social events for the School of Computer Science and Engineering
- **Volunteer at Ekta Nagar Residents Welfare Association** 2017 - 2018
Aided in the organisation of community activities and administrative affairs of my residential society
- **Organising Committee Member at Quark (BITS - Pilani Goa Technical Festival)** 2015
Directed the photo and video coverage of the technical festival spanned over 3 days
- **Videography intern at Zone Startups India** June 2015
Created an advertisement video, and clicked portfolio photographs for Zone Startups India
- **Member at The Department of Photography, BITS-Pilani Goa** 2012 - 2015
Performed photo and video coverage of campus events over 3 years and mentored new inductees